

AYMANE AIT DADS

AI Intern - Data Analysis & Engineering | Machine Learning · Data Analytics · Business Intelligence
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PROFESSIONAL SUMMARY

Final-semester Data Science engineering student who delivered a production-grade machine learning pipeline at Orange Maroc classifying network quality across 100K+ records and reducing manual analysis time by 60%, directly informing commercial service-tier decisions. Ranked 1st in EURECOM class for AI-generated image detection (0.791 AUC) and placed top 10% on a \$106K+ Kaggle competition, demonstrating consistent delivery of analytical and innovative data solutions. Core competencies span Python, SQL, Power BI dashboards, PyTorch, Scikit-learn, Hugging Face Transformers, and end-to-end model development with actionable stakeholder reporting. Available for a 6-month AI internship in Asia immediately; no sponsorship constraint as a French engineering-school graduate with international mobility.

CORE COMPETENCIES

Machine Learning | Data Analysis | Business Intelligence | AI Engineering | Analytical Problem-Solving | Model Evaluation | Data Storytelling | Prototype Development

WORK EXPERIENCE

Orange Maroc Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Developed a clustering pipeline mapping **100K+ fiber-optic network records to 5 commercial service tiers**, output adopted directly by the network optimization team for maintenance planning.
- Built Random Forest and K-Means models on upload speed, latency, and jitter features, reducing **manual analysis time by ~60%** through automated feature engineering and model evaluation workflows.
- Produced **Power BI dashboards** translating model outputs into executive-ready maintenance recommendations, enabling data-driven decisions across thousands of network nodes.
- Delivered stakeholder presentations distilling complex clustering results into **actionable business insights**, bridging technical model outputs and commercial decision-making for non-technical audiences.

PROJECTS

Aerial Cactus Detection - Endangered Species Classification EURECOM Deep Learning · 2025 · 99.8% Accuracy

- Benchmarked CNN, DenseNet121, and hybrid CNN-Transformer architectures on **17,500+ drone images**, achieving 99.8% accuracy, F1 = 0.999, and ROC AUC = 1.0 for endangered species classification.
- Conducted systematic model evaluation and ablation studies comparing backbone architectures, delivering **quantitative performance metrics** and visualizations that translated directly into deployment recommendations.

PyTorch · DenseNet121 · Scikit-learn · Matplotlib

Twitter Sentiment Analysis - End-to-End NLP Pipeline EURECOM NLP · 2025 · Business Sentiment Classification

- Built an end-to-end NLP pipeline progressing from tokenization and TF-IDF through word2vec embeddings to **transformer fine-tuning**, covering the full spectrum of text preprocessing for sentiment classification.
- Applied systematic hyperparameter optimization across model variants, generating **actionable insights on classification performance** and demonstrating data storytelling skills translating NLP outputs into business-relevant findings.

Python · Hugging Face Transformers · Scikit-learn · Pandas

NVIDIA Nemotron Reasoning Challenge - Kaggle Competition Kaggle · In Progress Jun 2026 · Top 10% · \$106K+ Prize Pool

- Achieved **top 10% public leaderboard score (0.80/1.0)** on a \$106K+ prize-pool competition by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) on 7,828 chain-of-thought reasoning examples.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict dependency order and ran 20+ single-variable ablations (packing, loss masking, data synthesis) to isolate completion-only loss as the key accuracy driver.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya 2021 - 2023

SKILLS

Data Science & Analytics: Python | SQL | Pandas | NumPy | Power BI | Feature Engineering | Scikit-learn

Machine Learning & AI Engineering: PyTorch | Hugging Face Transformers | LoRA / PEFT | Computer Vision | NLP | Ensemble Methods

MLOps & Infrastructure: Git | Linux | FP16 Mixed Precision | Kaggle GPU | REST APIs | Docker (basic) | PostgreSQL